MA 420G · Spring 2027

Introduction to Stochastic Processes

The mathematics behind randomness

TIME AND ROOM (TENTATIVE)
MWF 9:00–9:50 am · CB 203

INSTRUCTOR
Tianshi Xu

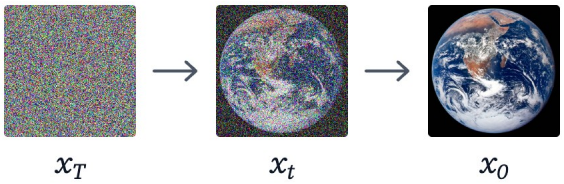
CREDITS
3

PREREQUISITE
MA/STA 320 or STA 524; or graduate student status; or consent of the department.

COURSE DESCRIPTION

Stochastic processes describe systems that change randomly over time. This course develops the mathematics of Markov chains, Poisson processes, martingales, and Brownian motion, connecting theory with applications to long-run behavior, waiting times, and decisions under uncertainty.

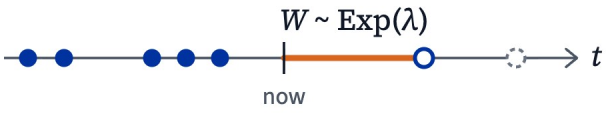
QUESTIONS WE WILL EXPLORE



How can noise become an image?

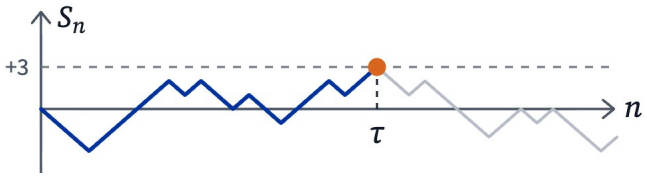
Diffusion models turn noise into images. **Markov chains** help us understand the mathematics behind this process.

Photo: NASA Johnson Space Center / Apollo 17 (1972).



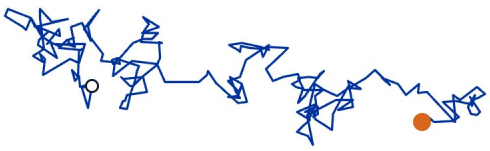
How long until the next arrival?

Messages and customers arrive at unpredictable times. **Poisson processes** and exponential distributions describe random arrivals and waiting times.



Can a stopping rule beat a fair game?

Random walks model the ups and downs of a fair game. **Martingales** help us understand expected rewards and the effect of stopping rules.



Why does pollen keep moving?

Countless molecular collisions make tiny particles wander. **Brownian motion** provides a mathematical model for these irregular paths.